Dr. Son Tong

Senior Research Engineer

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R&D Engineer and Manager in Control Systems, Robotics, ADAS and Autonomous Vehicle

Research and Education

2016 - now **Senior Researcher**, Siemens Digital Industries Software, Leuven, Belgium.
Research and commercial projects execution, R&D ADAS team manager, collaboration with industry and academic partners

Oct. 2016 **Ph.D. in Mechanical Engineering**, KULeuven, Leuven, Belgium.
Advisors: Prof. Jan Swevers, Prof. Goele Pipeleers

Feb. 2012 **M.S. of Research in Mechatronics**, GIST, Gwangju, South Korea.
Advisor: Prof. Hyo-Sung Ahn

Jun. 2009 **B.S. in Electrical Engineering**, HUST, Hanoi, Vietnam.

Invited Presentations

Oct. 2021 **Invited industrial speaker in Politechnico Milano**, Italy.
To the Automation and Control Engineering Degree Programme (more than 400 BSc and MSc students)

May 2021 **Oxford Control Group**, UK.
https://eng.ox.ac.uk/control/seminars/

March 2021 **HyConSys Lab, TU Munich**, Germany.
https://www.hyconsys.com/talks.html

April 2021 **IROHMS, Cardiff University**, UK.
https://www.cardiff.ac.uk/artificial-intelligence-robotics-and-human-machine-systems

May 2019 **Hori-Fujimoto Lab, University of Tokyo**, Japan.
https://sites.google.com/edu.k.u-tokyo.ac.jp/hflab/home

May 2018 **Automatic Control Laboratory, EPFL**, Lausanne, Switzerland.
https://memento.epfl.ch/event/simulation-based-testing-and-validation-framework-/

Sept. 2018 **AutoSens Conference**, Brussels, Belgium.
AutoSens is an autonomous vehicle event, participants are from the ADAS sensing industry

Projects and Grants

Commercial projects **Siemens Engineering Services**, Support and execute commercial (and confidential) engineering projects in automotive industry for OEMs and suppliers.

EU Projects **EU Horizon 2020 FOCETA**, FOundations for Continuous Engineering of Trustworthy Autonomy, http://www.foceta-project.eu/.
I am the technical coordinator and project manager from Siemens with multiple involved departments

EU Horizon 2020 Enable-S3, European Initiative to Enable Validation for Highly Automated Safe and Secure Systems, https://enable-s3.eu/.
I am the Siemens technical coordinator of the ADAS work package on autonomous valet parking

Marie Curie Innovative Training Network ELO-X, Embedded learning and optimization for the next generation of smart industrial control systems, https://elo-x.eu/.
I am in Scientific Advisors Board and the co-supervisors of several Marie Curie (PhD) Fellows

Belgium **Flemish Research Schemes**.
I'm project manager of multiple projects (mostly on ADAS and control systems), and Siemens coordinator in the Flanders Make Decision and Control cluster.

- EMDAS (2015 - 2107): Environmental Modelling for automated Driving and Active Safety
- ROFALC (2016 - 2018): Robust and Fast Learning Control
- AVCON (2017 - 2019): Avoidance of collisions and obstacles in narrow lanes
- CONACON (2019 - 2021): Context Aware Control
- ID2CON (2020 - 2022): Integrated IDentification for CONtrol
- MARLOC (2020 - 2022): Mixed absolute and relative localization

Awards

Jan. 2019 **2018 Siemens DF PL Invention of the Year Award**.
This is a yearly competition to recognize and award the most significant invention during the fiscal year from the Siemens DF PL business unit worldwide. The award includes cash prize and company announcement

Sept. 2019 **Finalist of the AutoSens Award 2019 in Most Influential Research**.
https://auto-sens.com/awards-2019-finalists-revealed/

2012-2015 **Marie Curie Early Stage Researcher Fellowship**.
EU research and innovation fellowships programme for doctoral education and training

2011 **American Control Conference**.
Best Presentation in Session Award

2010-2012 GIST Scholarship for Master student

2009 International Internship Scholarship in South Korea

2005 HUST scholarship for exellent students

Experience

2016–Present **Senior Research Engineer**, Siemens.
Working on multiple European and Belgian research and development projects, with focus on control and ADAS and autonomous vehicle. Setup collaborations with academic institutes.

- Autonomous valet parking, highway pilot, green wave technology, intersection crossing
- ADAS safety and comfort
- Model predictive control (MPC) developments with real-time implementations
- Physical vehicles and simulation testing frameworks
- AI algorithms: deep learning, Gaussian process, imitation learning, reinforcement learning
- Driving style classification
- Scenario generation

2016–Present **Project Management and Supervisor**, Siemens.

- Manage control and ADAS/AV projects: coordination, proposal writing
- Supervise PhD, researcher, Master, and intern students in the R&D team
- Develop and manage autonomous driving vehicle platforms

2012–2015 **EU Marie Curie ITN Training Programme**, EU FP7 IMESCON Project.

- EU Marie Curie scholarship to do PhD and training programs
- Model identification and control design for the amplifed piezo actuator, Cedrat Technologies (France).

2012–2016 **Advance Control Design for Mechatronic Systems**, KU Leuven, Belgium.

- Control theory: linear and nonlinear control, robust control, optimal control...
- Design techniques: PID, loop-shaping, model-based control, H-infinity control, MIMO control, feedforward control, model predictive control (MPC)...
- Optimization tools: convex optimization, LMI, optimization softwares
- Experimentally validated on a lab-scale overhead crane and XY wafer stage setups.

2012–2016 **Iterative Learning Control (ILC)**, KU Leuven, Belgium.

- Main PhD research topic
- Proposed a novel robust norm-optimal iterative learning control in time domain
- Proposed and designed a multi-objective ILC problem in frequency domain
- Developed multivariable ILC analysis and design

2010–2012 **Research Assistant**, GIST, South Korea.

- Did research in the Distributed Control and Autonomous Systems Lab. (GIST)
- Developed multiple points tracking iterative learning control

Professional Services

Editorial IEEE Control System Society Conference Editorial Board since 2020 (for ACC and CDC conferences)

Technical Committee On Automotive Control and Smart Cities Control

Reviewer Transport Research Arena, IEEE Transaction on Automatic Control, IEEE Transaction on Control System Technology, International Journal of Robust and Nonlinear Control, Control & System Letters, Mechatronics, IEEE Conference on Decision and Control, American Control Conference, IFAC conferences

Teaching and Supervision

2016- **Siemens R&D ADAS Team Manager** of research engineers and industrial PhDs

2016- **Supervising PhD and Master thesis students** from KULeuven, EPFL, Linkoing Univ., PoliTo, UTokyo, Grenoble INP, Delft Univ...

2012-2015 **Master course in KULeuven:** Control Theory Exercise Sessions

2012-2015 **Master course in KULeuven:** System Theory Exercise Sessions

Publications

The complete list can be found on my Google Scholar page

1. Allamaa JP., Listov P., Van der Auweraer H., Jones C., **Son, T.D.**, "Real-time Nonlinear MPC Strategy with Full Vehicle Validation for Autonomous Driving", ***arXiv:2110.03349***, 2021
2. Acerbo FS., Alirczaei M., Van der Auweraer H., **Son, T.D.**, "Safe Imitation Learning on Real-Life Highway Data for Human-like Autonomous Driving", *IEEE International Intelligent Transportation Systems Conference (ITSC)*, Indianapolis, Sept. 2021
3. Bruggner D., Hegde A., Acerbo FS., Gulati D., **Son, T.D.**, "Model in the Loop Testing and Validation of Embedded Autonomous Driving Algorithms", *32nd IEEE Intelligent Vehicles Symposium*, Nagoya, July 2021
4. **Son, T.D.**, Lanh, N., and Van der Auwerier H., "Learning control applications for autonomous driving in extreme maneuver scenarios", *The International Conference on Control, Automation, and Systems*, Jeju, Oct. 2021

5. Diwale S., **Son, T.D.**, Jones C., “Manifold learning and optimal control for obstacle avoidance in autonomous driving”, *submitted*
6. Arany RR., Van der Auweraer H., **Son, T.D.**, “Learning Control for Autonomous Driving on Slippery Snowy Road Conditions’, *IEEE Conference on Control Technology and Applications (CCTA)*, Montreal,Sept. Sept. 2020
7. Kneissl M., vom Dorff S., Molin A., Denniel M., **Son, T.D.**, Ochoa Lleras N., Esen H., Hirche S., “Mixed-Reality Testing of Multi-Vehicle Coordination in an Automated Valet Parking Environment”, *The 21th World Congress IFAC*, Berlin, Aug. 2020
8. Acerbo FS., Van der Auweraer H., **Son, T.D.**, “Safe and Computational Efficient Imitation Learning for Autonomous Vehicle Driving”, *American Control Conference (ACC)*, Denver, July 2020
9. **Son, T.D.**, Quan N., “Safety-critical control for non-affine nonlinear systems with application on autonomous vehicle”, *58th IEEE Conference on Decision and Control*, Nice, Dec. 2019
10. **Son, T.D.**, Bhave A., and Van der Auwerier H., “Simulation-based testing framework for autonomous driving development”, *IEEE 2019 International Conference on Mechatronics*, Ilmenau, Mar. 2019
11. **Son, T.D.**, Awatsu L., Hubrechts J., Bhave A., and Van der Auwerier H., “A simulation-based verification and testing framework for ADAS development”, ***Transport Research Arena***, Vienna, 2018
12. Steinhauser, A., **Son, T.D.**, Hostens, E., and Swevers, J., “ROFALT: An Optimization-based Learning Control Tool for Nonlinear Systems", *The 15th International Workshop on Advanced Motion Control*, Tokyo, Mar. 2018
13. **Son, T.D.**, Pipeleers, G., and Swevers, J., “Multi-objectivei iterative learning control using convex optimization”, ***European Journal of Control***, Jan. 2017
14. **Son, T.D.**, Pipeleers, G., and Swevers, J., “Robust monotonic convergent iterative learning control”, ***IEEE Transactions on Automatic Control***, Issue 99, Jul. 2015
15. **Son, T.D.**, Ahn, H.S., and Moore, K., “ Iterative learning control in optimal tracking problems with specified data points”, ***Automatica***, Issue 5, May 2013
16. **Son, T.D.**, A., Pipeleers, G., and Swevers, J., and Van der Auwerier H., “A Generalized Frequency Domain Learning Control Design with Experimental Validation", *The 43rd Annual Conference of the IEEE Industrial Electronics Society*, Beijing, Nov. 2017
17. **Son, T.D.**, Steinhauser, A., Pipeleers, G., and Swevers, J., “Robust performance iterative learning control : Analysis, synthesis and experimental validation”, *The European Control Conference (ECC16)*, Denmark, Jul. 2016
18. **Son, T.D.**, Pipeleers, G., and Swevers, J., “Robust analysis and synthesis with unstructured model uncertainty in lifted system iterative learning control”, *2015 American Control Conference (ACC15)*, Chicago, Jun. 2015
19. **Son, T.D.**, Pipeleers, G., and Swevers, J., “Experimental validation of robust iterative learning control on an overhead crane test setup”, *The 19th World Congress IFAC 2014*, Cape Town, Aug. 2014
20. **Son, T.D.**, Pipeleers, G., and Swevers, J., “Robust optimal iterative learning control with model uncertainty”, *The 52nd IEEE Conference on Decision and Control (CDC13)*, Florence=, Dec. 2013
21. **Son, T.D.**, Pipeleers, G., and Swevers, J., “Optimal iterative learning control design with trial-varying initial conditions”, *The European Control Conference (ECC13)*, Zurich, Jul. 2013
22. **Son, T.D.**, Ahn, H.S., “Optimal iterative learning control with uncertain reference points”, *The 2012 IEEE Multi-Conference on Systems and Control*, Dubrovnik, Oct. 2012

23. **Son, T.D.**, Ahn, H.S., "Iterative learning control for optimal multiple-point tracking", *The 50th IEEE Conference on Decision and Control and European Control Conference(CDC-ECC 2011)*, Orlando, Dec. 2011
24. **Son, T.D.**, Ahn, H.S., "An interpolation method of multiple terminal iterative learning control", *The 2011 IEEE Multi-Conference on Systems and Control (MSC 2011)*, Denver, Sept.. 2011
25. **Son, T.D.**, Ahn, H.S., "Terminal iterative learning control with multiple intermediate pass points", *The 2011 American Control Conference (ACC11)*, San Francisco, Jun. 2011